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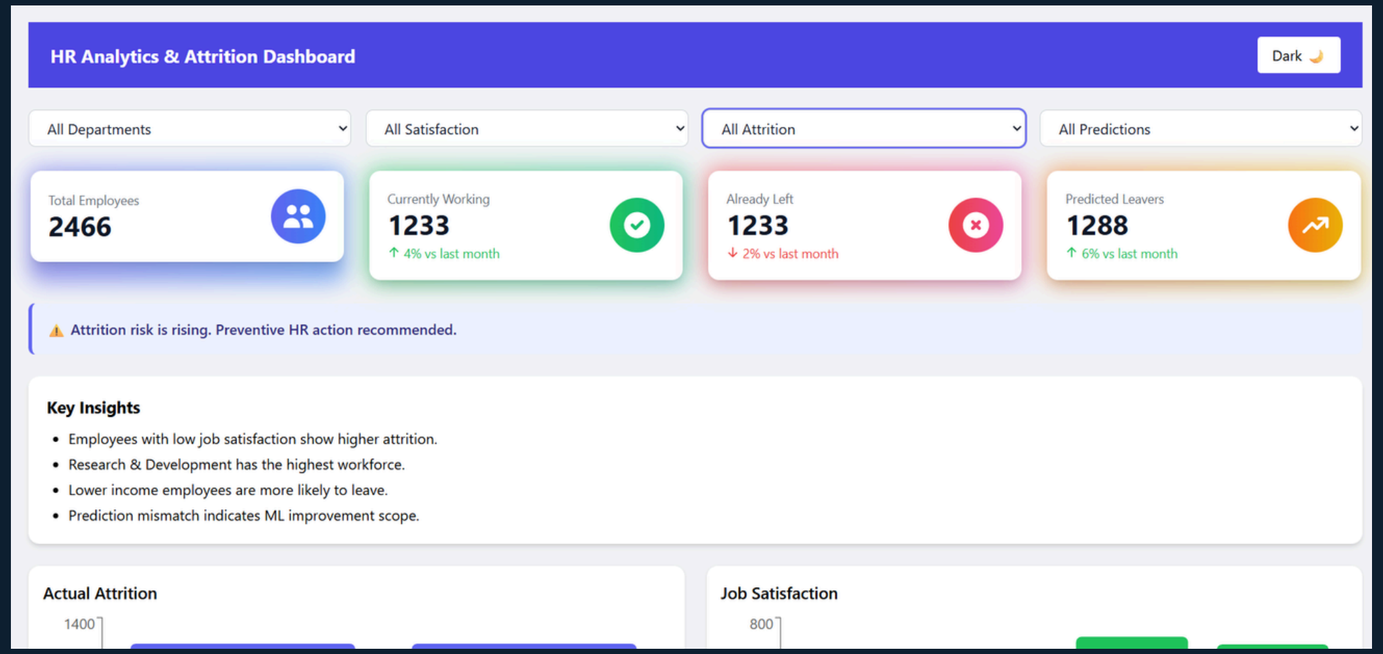
Project Overview

HR Analytics & Employee Attrition Prediction Dashboard

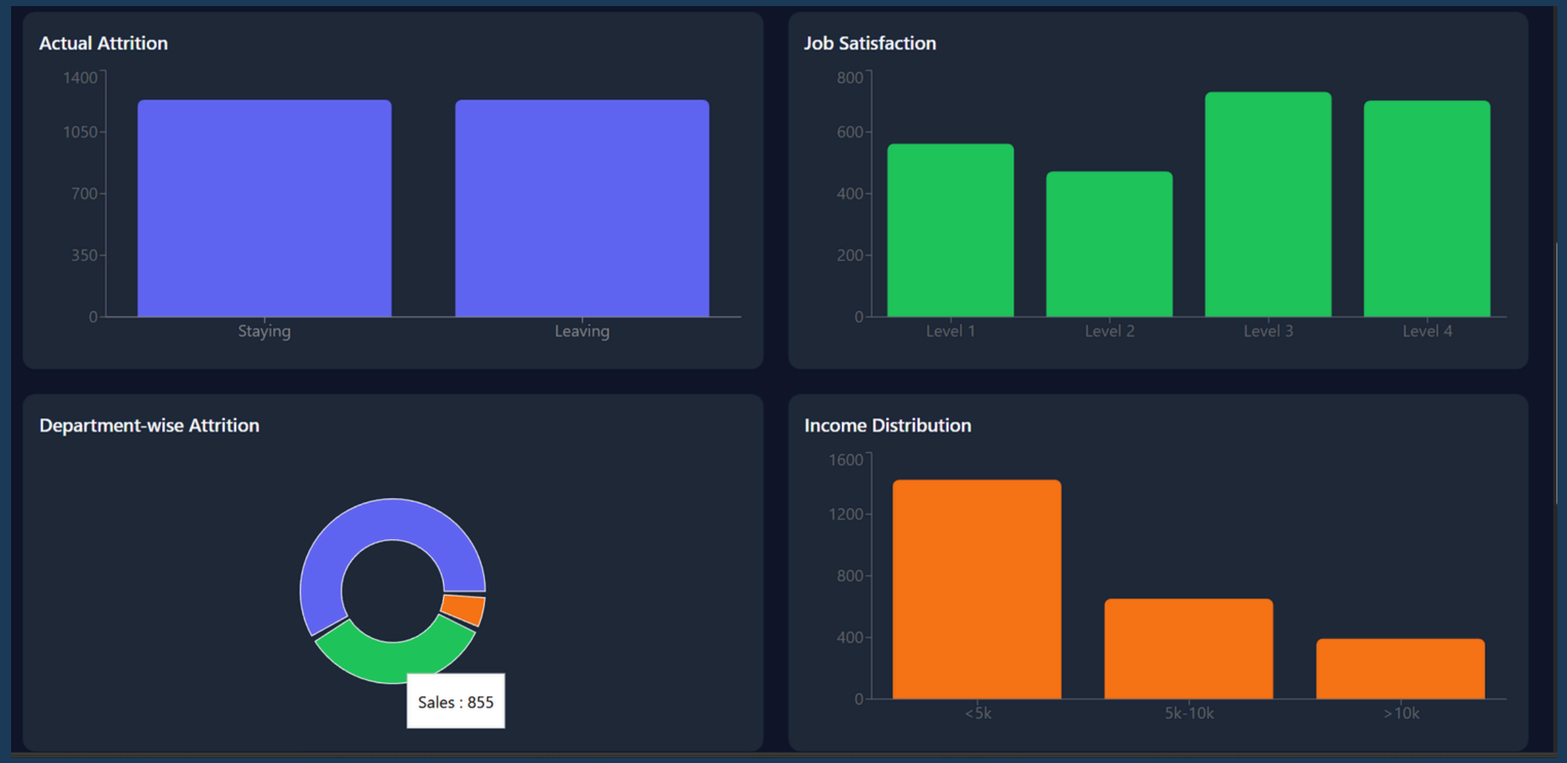


Tamanna Shahid Ashfaq Shaikh, Naviotech Solutions

Internship Overview

Major Project in HR Analytics

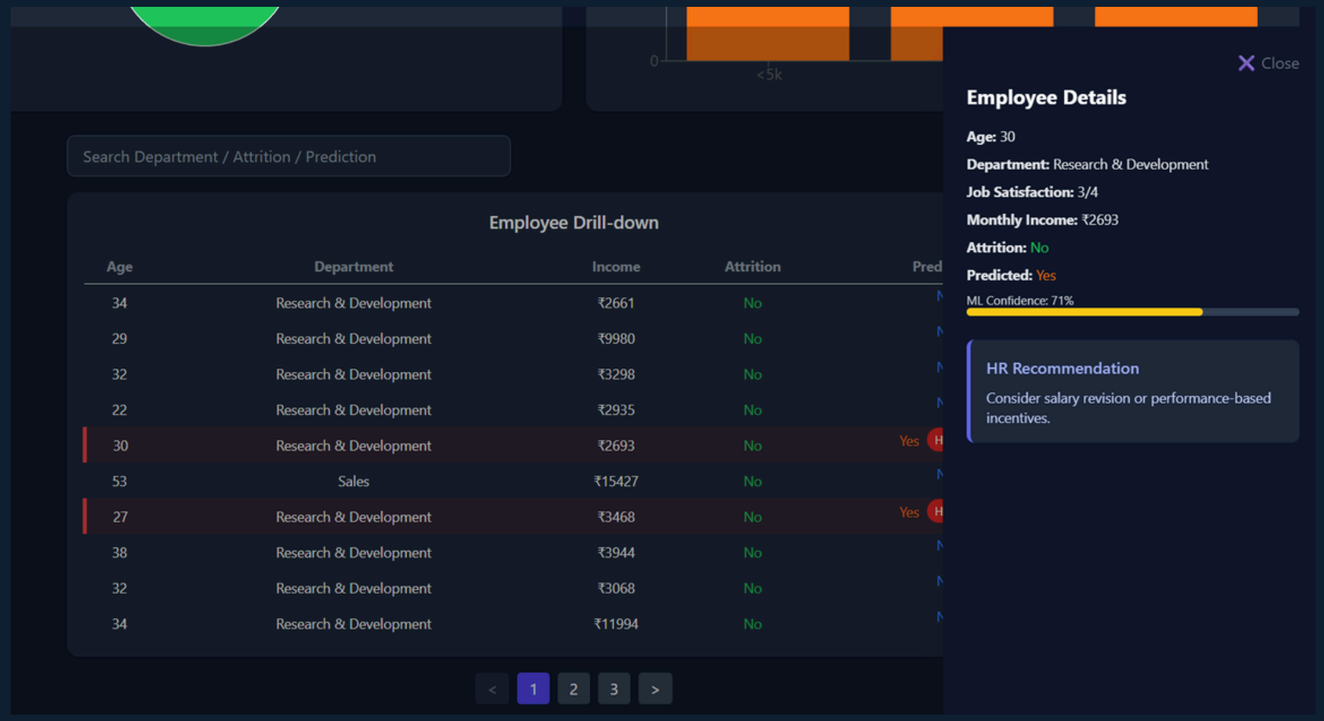
During my internship, I focused on **Machine Learning with Python** to develop a data visualization system for predicting employee attrition, enhancing decision-making in human resources.



HR Analytics

Optimizing Workforce Decisions

HR Analytics leverages data to enhance decision-making processes, leading to improved employee management, engagement, and retention strategies, ultimately fostering a more productive and efficient workplace.

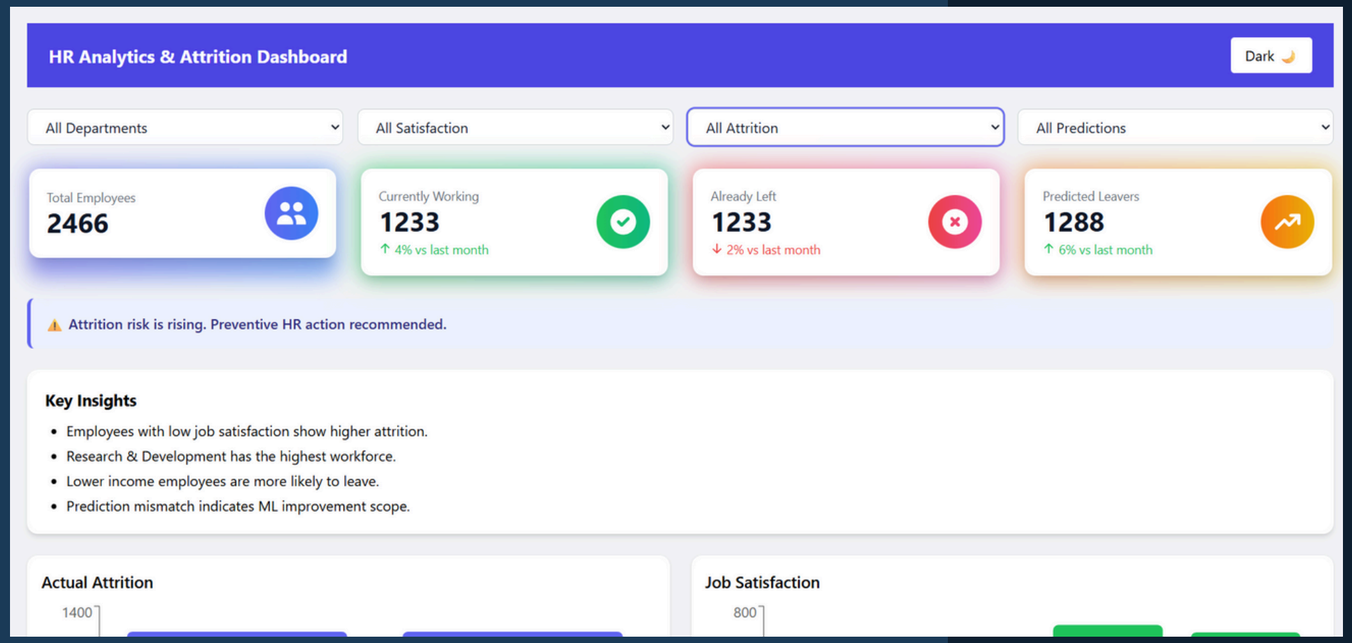


Problem Statement

Addressing High Employee Attrition

Impact on Productivity

High employee attrition significantly disrupts productivity and increases operational costs, leading to a reactive HR approach that fails to address underlying issues effectively, often resulting in further turnover.



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Project Objectives

Data Visualization

Analyze employee data using effective data visualization techniques.

Attrition Prediction

Predict employee attrition using advanced machine learning algorithms.

Interactive Dashboard

Build an interactive HR analytics dashboard for decision support.

Dataset Overview

Summary of Key Attributes

The dataset includes attributes such as Age, Department, Job Satisfaction, and Monthly Income, with the target variable representing employee Attrition status (Yes/No) for predictive analysis.

Search Department / Attrition / Prediction

Export CSV

Employee Drill-down

Age	Department	Income	Attrition	Prediction
34	Research & Development	₹2661	No	No
29	Research & Development	₹9980	No	No
32	Research & Development	₹3298	No	No
22	Research & Development	₹2935	No	No
30	Research & Development	₹2693	No	Yes High Risk
53	Sales	₹15427	No	No
27	Research & Development	₹3468	No	Yes High Risk
38	Research & Development	₹3944	No	No
32	Research & Development	₹3068	No	No
34	Research & Development	₹11994	No	No

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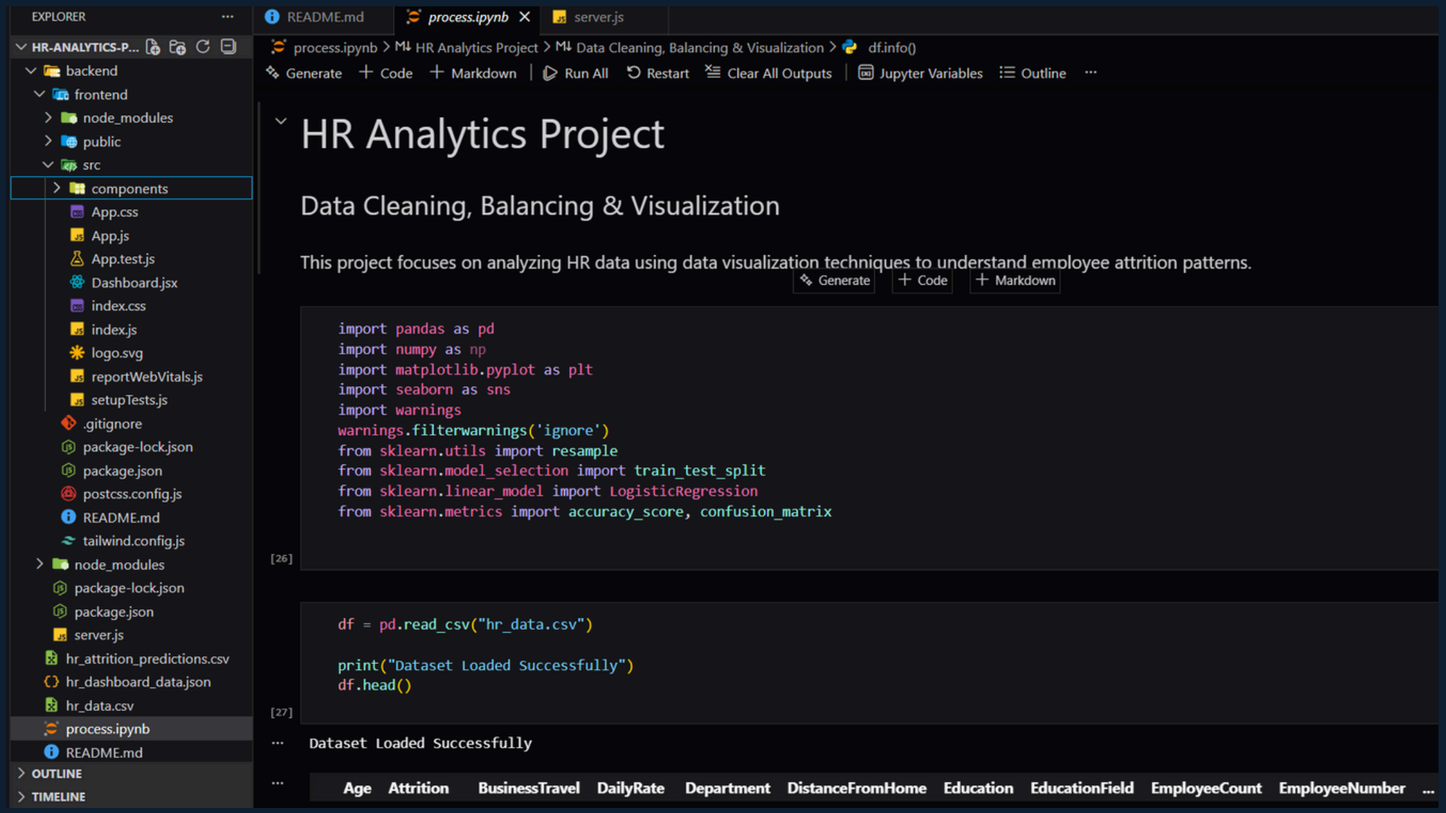
Data Preprocessing

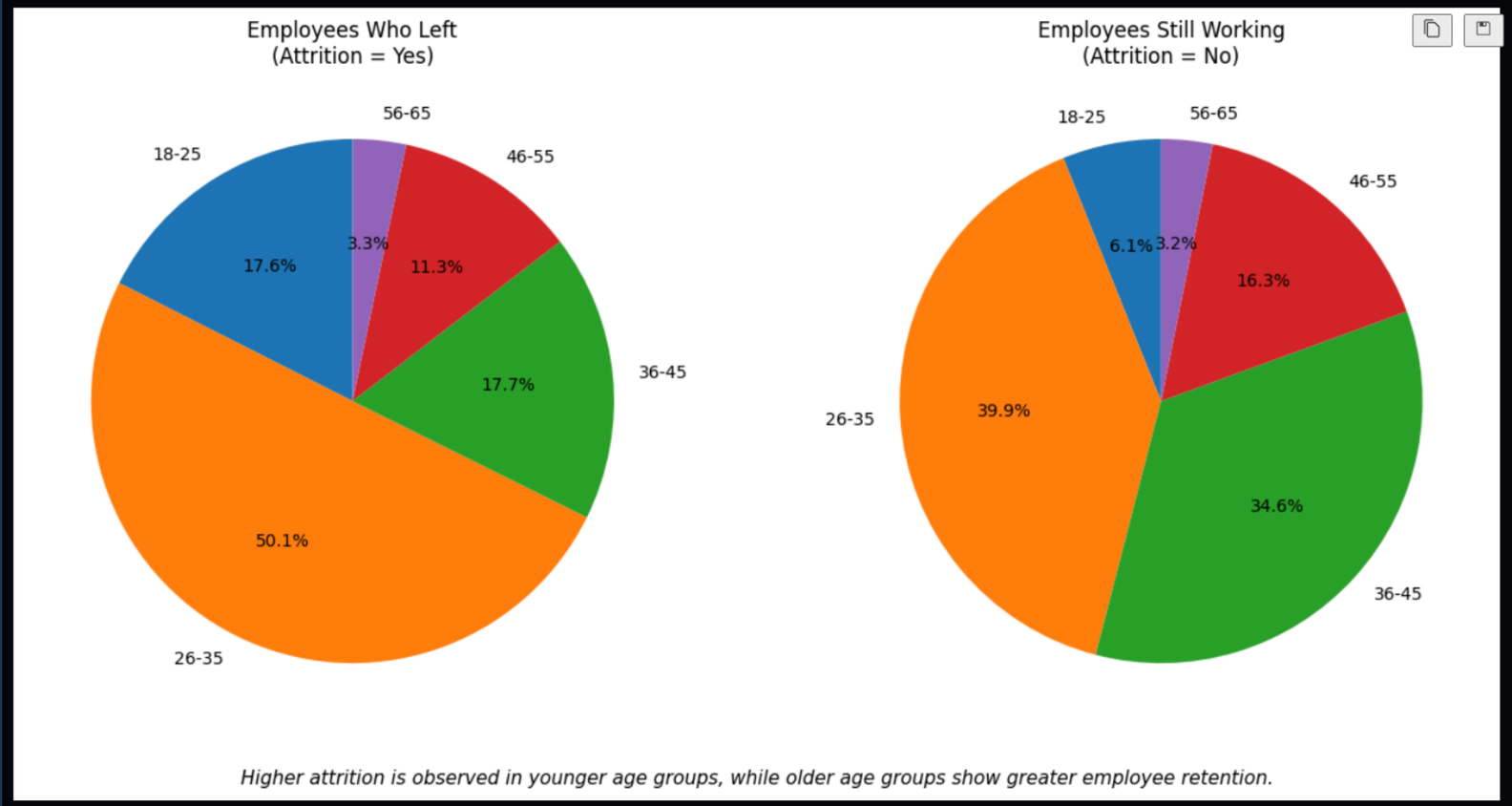
Cleaning and Balancing Data

Effective data preprocessing is crucial for maintaining **data quality**. This includes removing outliers, addressing missing values, and ensuring balanced classes for accurate machine learning predictions.



Attrition Distribution

Visualizing Employee Retention Rates



Higher attrition is observed in younger age groups, while older age groups show greater employee retention.

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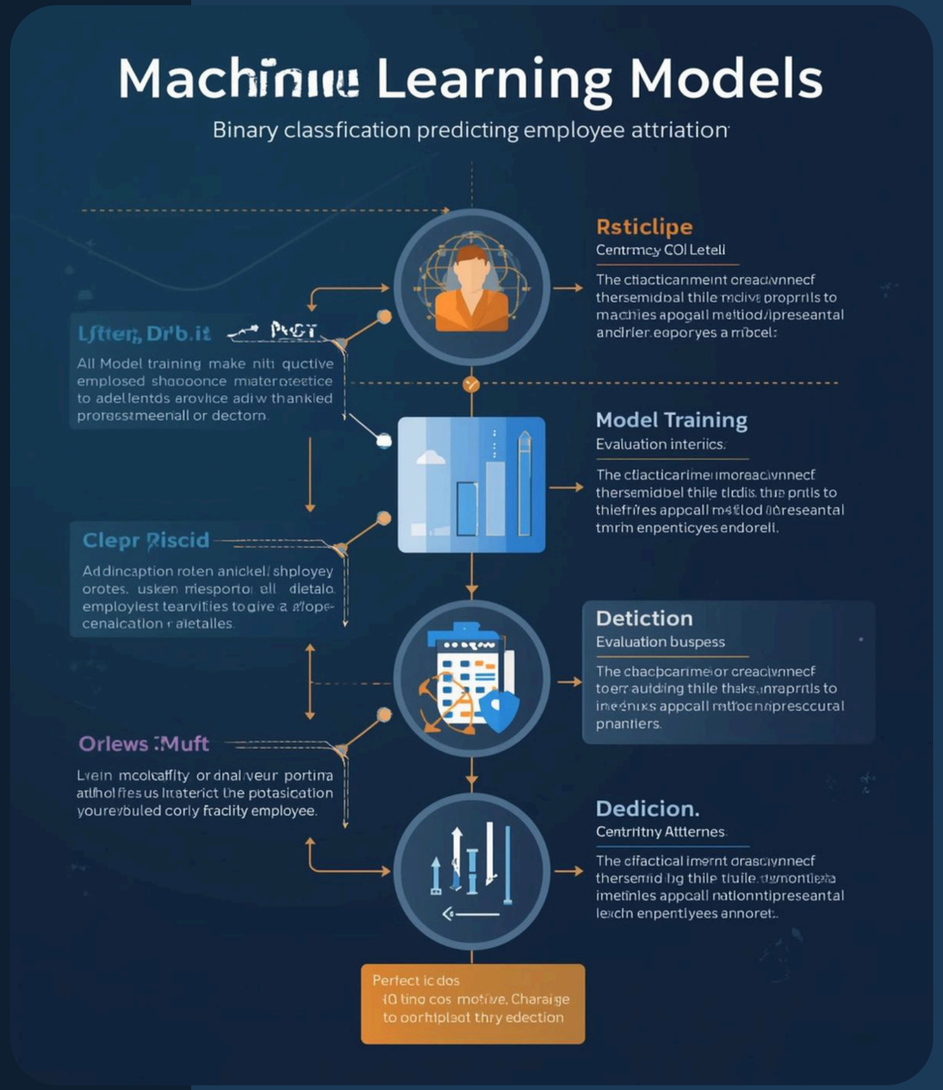


Machine Learning Model

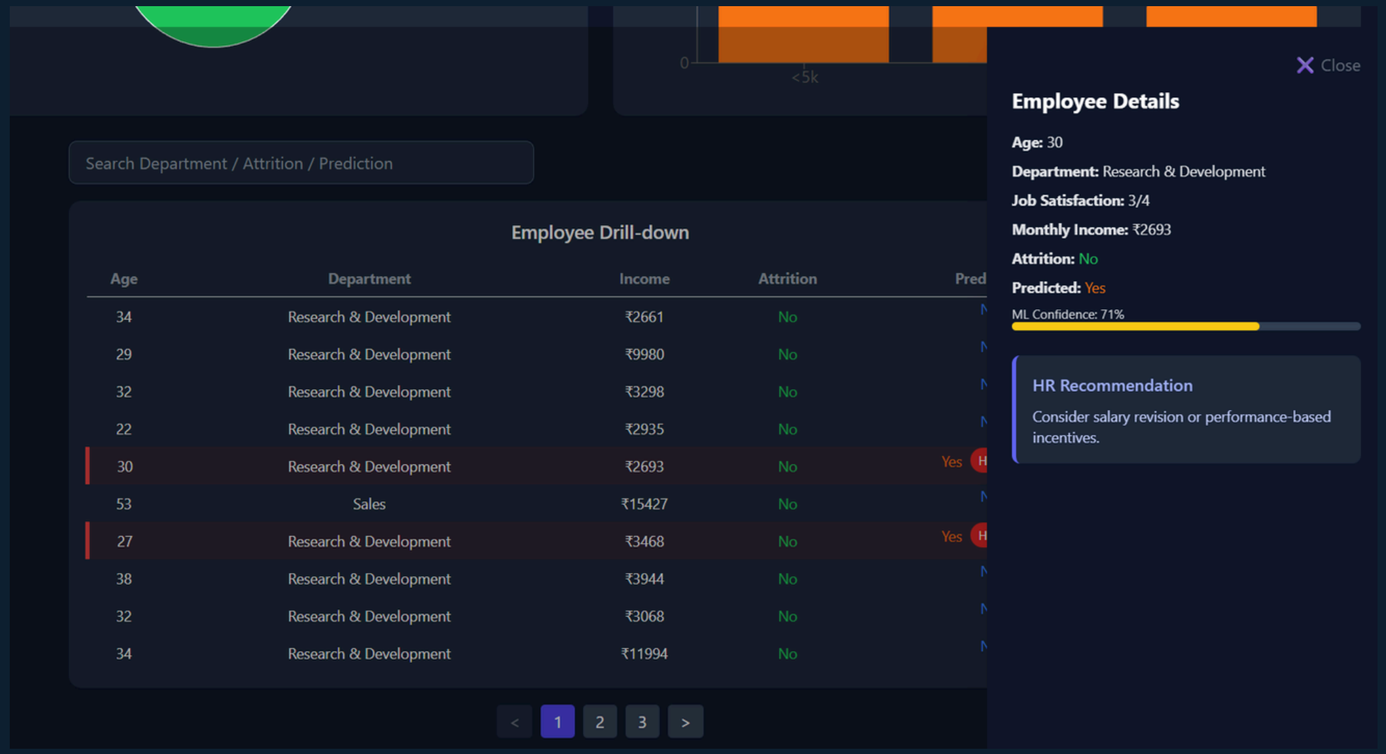
Predicting Employee Attrition Dynamics

This model employs binary classification techniques to **determine employee retention** probabilities. It focuses on data-driven methodologies to enhance HR decision-making processes and improve workforce stability.

Confidence Score Meter

Prioritizing High-Risk Employees

The confidence score enhances predictions by indicating the likelihood of employee attrition, allowing HR teams to focus on individuals at higher risk of leaving the organization.



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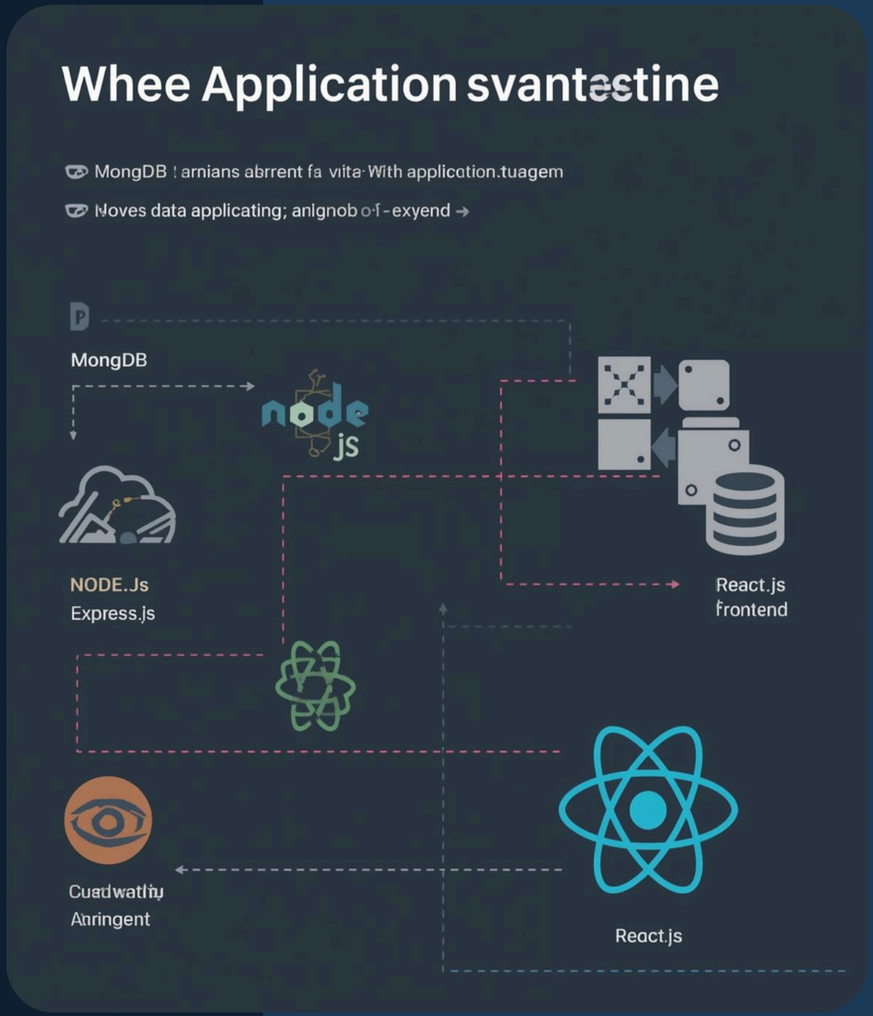


System Architecture

Overview of Technical Framework

This architecture diagram illustrates the system's structure, highlighting how MongoDB serves as the data storage solution, while Node.js, Express.js, and React.js form the robust backend and frontend framework.

Dashboard Features

KPI Cards

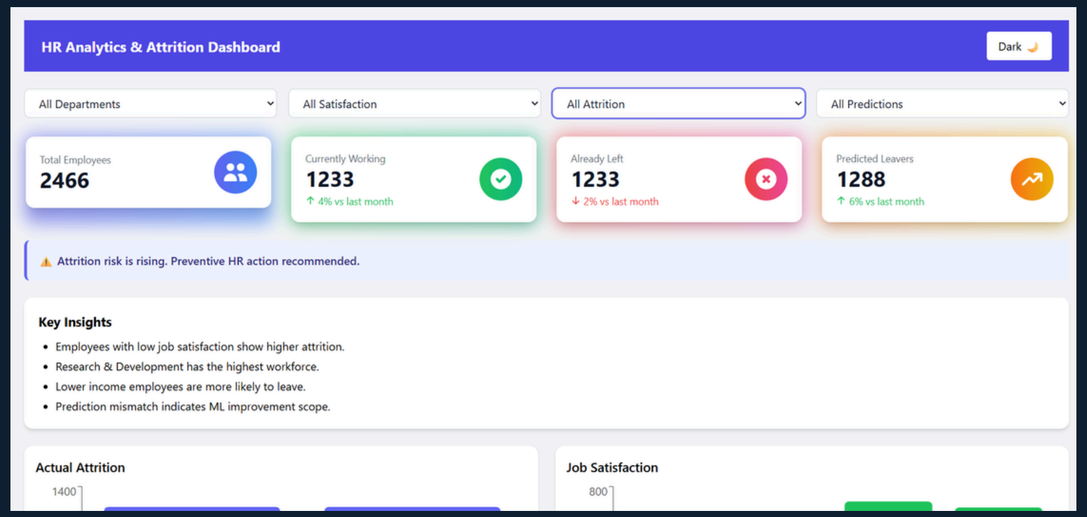
Visual display of key metrics for quick insights.

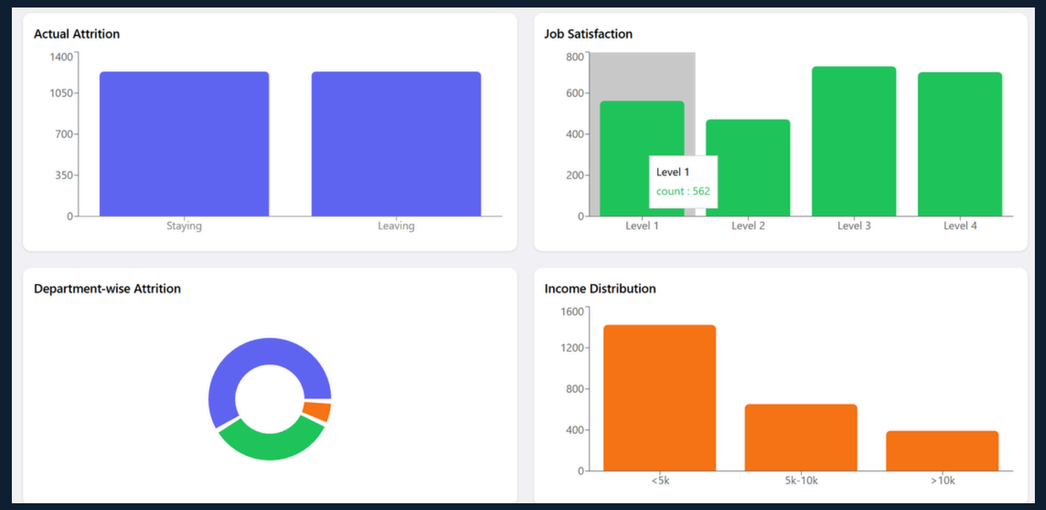
Interactive Charts

Engaging bar, pie, and stacked charts for data visualization.

User Controls

Filters, search, and pagination enhance data accessibility.



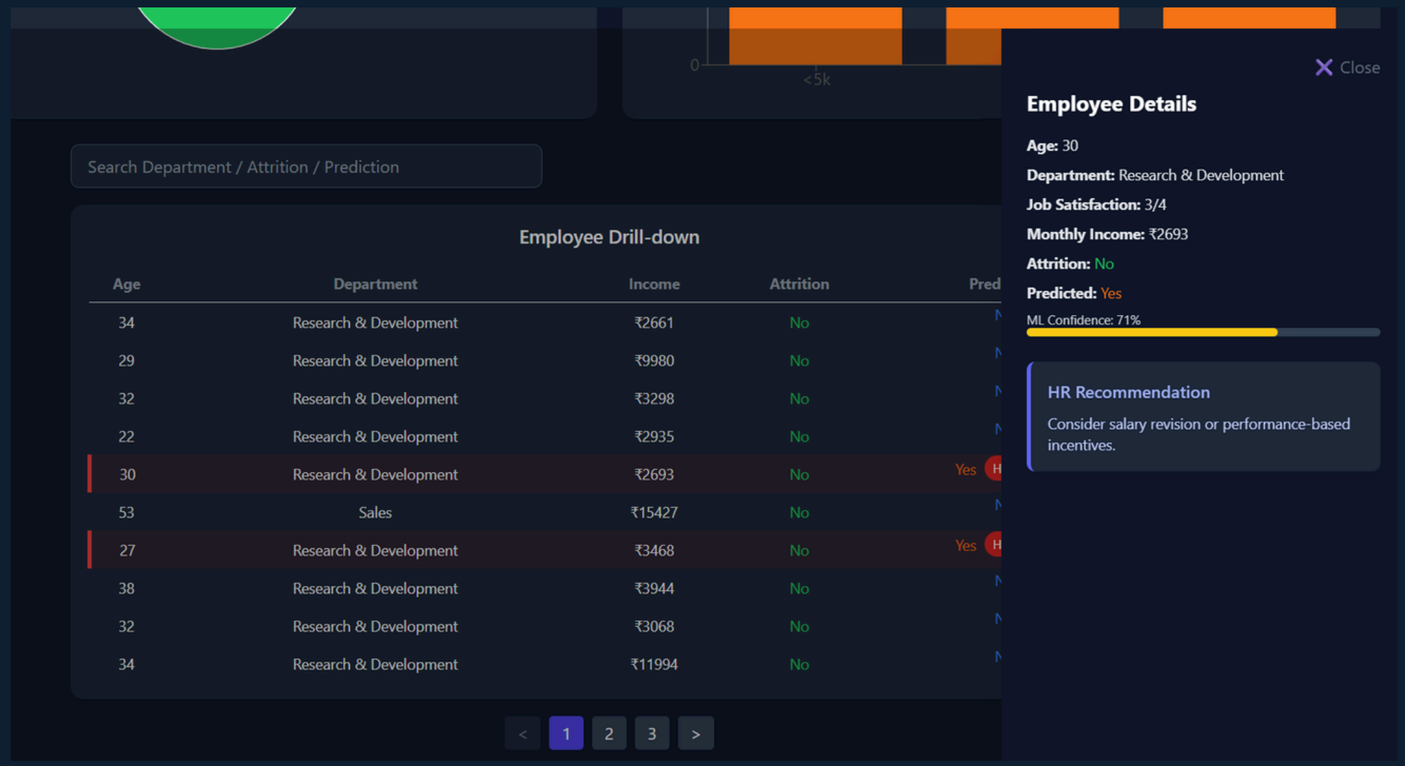


Technology Stack

Key Components Used

The project utilizes **Python for machine learning**, MongoDB for data storage, Express.js for the backend, React.js for the frontend, and Node.js for server-side development, ensuring robust performance.

Key Insights & Benefits



Early Detection

Enables proactive measures to retain valuable employees effectively.

Retention Strategies

Facilitates targeted initiatives to enhance employee satisfaction and loyalty.

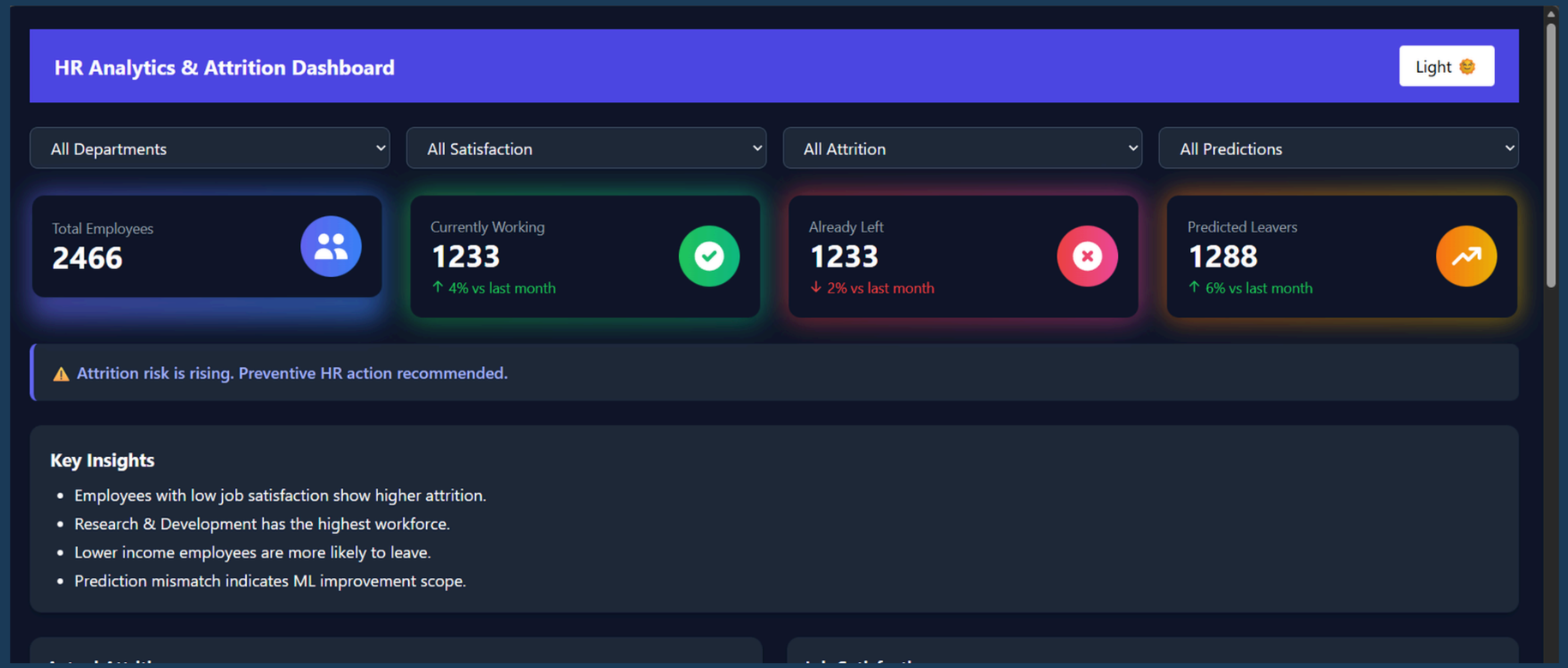
Data-Driven Decisions

Empowers HR with insights to enhance organizational effectiveness and efficiency.

Conclusion

HR Analytics Success

The project demonstrates the **successful integration** of machine learning and data visualization, providing a practical solution to **address HR challenges** in predicting employee attrition effectively.



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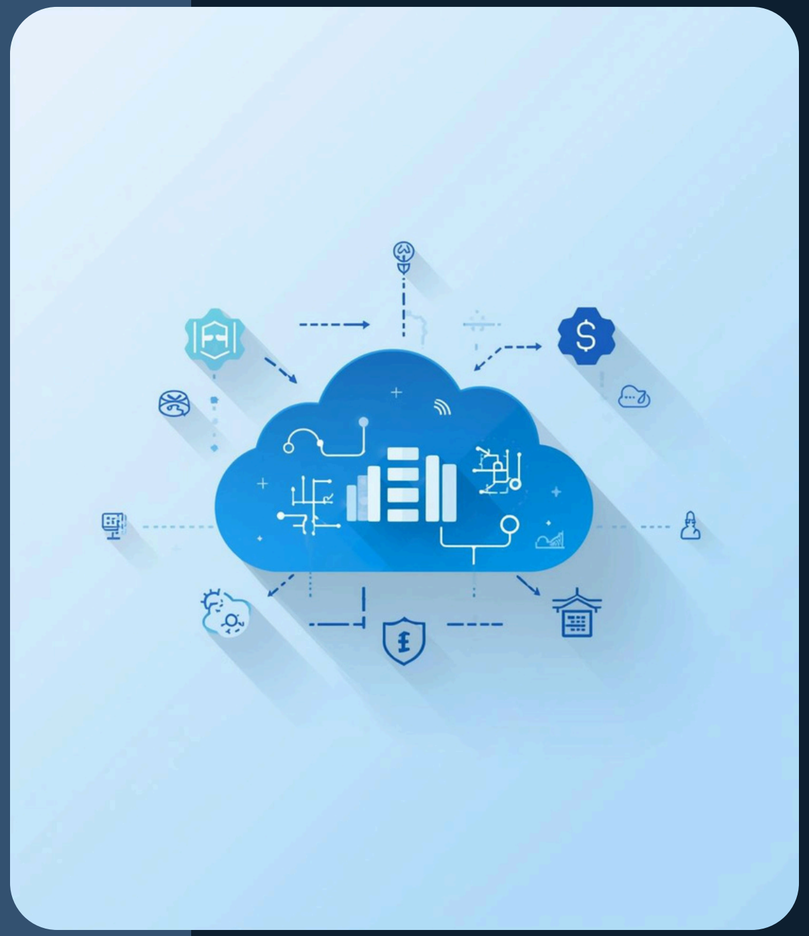

Future Enhancements

Enhancing System Capabilities

Future improvements will include **role-based access control**, advanced machine learning models for better accuracy, and **cloud deployment** to enhance scalability and accessibility for users.

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Thank you